

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **AYUSH MOHAPATRA 192411263** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

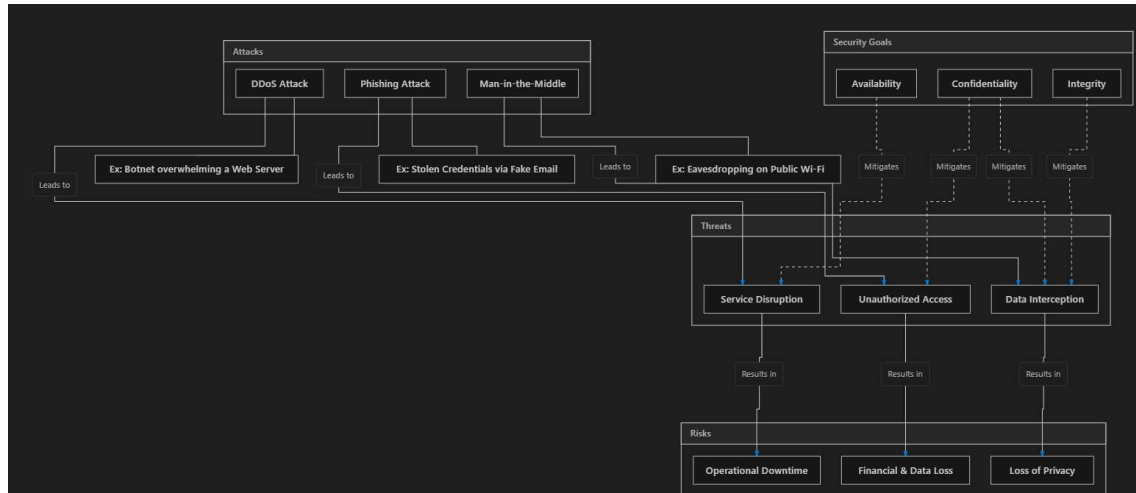
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Completed Assignment Answers

1. Security Attacks, Threats, Risks, and Security Goals



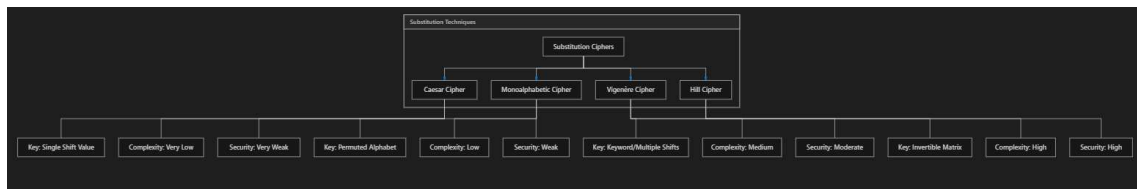
Concept Map

- Attacks: Phishing Attack, Man-in-the-Middle, DDoS Attack
- Leads to Threats: Unauthorized Access, Data Interception, Service Disruption
- Results in Risks: Financial & Data Loss, Loss of Privacy, Operational Downtime
- Mitigated by Security Goals: Confidentiality, Integrity, Availability

Analysis & Mitigation Mapping

- Phishing Attack leads to the threat of Unauthorized Access, which carries the risk of Financial & Data Loss. Confidentiality mechanisms (like Multi-Factor Authentication) help mitigate this threat. Real-world example: A deceptive email tricking an employee into revealing their password.
- Man-in-the-Middle (MitM) Attack leads to the threat of Data Interception/Modification, which carries the risk of Loss of Privacy or False Information. Confidentiality (Encryption) and Integrity (Hashing/Digital Signatures) mitigate this. Real-world example: An attacker intercepting unencrypted traffic on a public Wi-Fi network.
- DDoS Attack leads to the threat of Service Disruption, carrying the risk of Operational Downtime & Reputation Damage. Availability mechanisms (like Load Balancers and Redundancy) mitigate this threat. Real-world example: A botnet overwhelming a bank's servers, preventing users from accessing their accounts.

2. Substitution Techniques



Concept Map

- Caesar Cipher (Key: Single Shift Value, Complexity: Very Low, Security: Very Weak)
- Monoalphabetic Cipher (Key: Permuted Alphabet, Complexity: Low, Security: Weak)
- Vigenère Cipher (Key: Keyword/Multiple Shifts, Complexity: Medium, Security: Moderate)
- Hill Cipher (Key: Invertible Matrix, Complexity: High, Security: High)

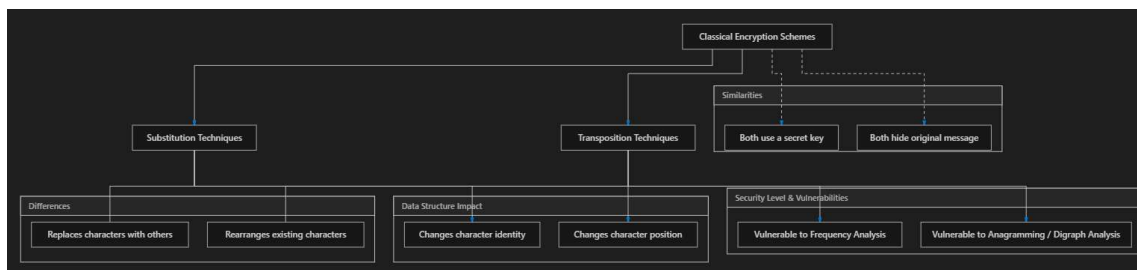
Analysis

Which technique is most secure and why?

The Hill Cipher (and arguably the Vigenère Cipher depending on key length) is the most secure among these classical techniques.

- Caesar and Monoalphabetic are simple substitution ciphers. They preserve the underlying language's letter frequencies exactly, making them highly vulnerable to Frequency Analysis.
- Vigenère Cipher is a polyalphabetic substitution cipher, which flattens the frequency distribution, making standard frequency analysis much harder. However, if the key length is discovered, it can be broken.
- Hill Cipher encrypts multiple letters (blocks) at a time using matrix multiplication. This diffuses the statistical properties of the plaintext across the block, making single-letter frequency analysis useless. Its high mathematical complexity and block-based approach make it the most secure of the given classical substitution techniques.

3. Classical Encryption Schemes



Concept Map

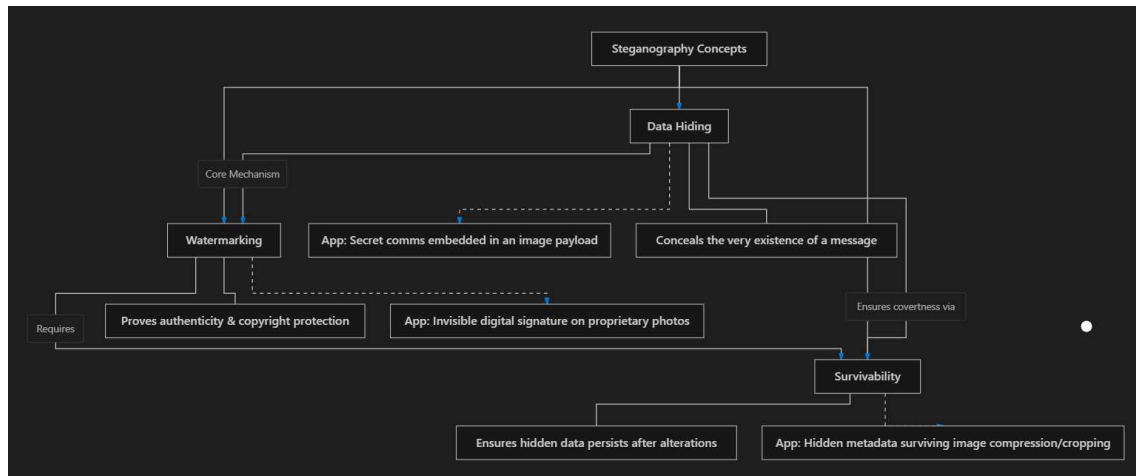
- Substitution Techniques: Replaces characters with others (Changes character identity). Vulnerable to Frequency Analysis.
- Transposition Techniques: Rearranges existing characters (Changes character position). Vulnerable to Anagramming / Digraph Analysis.
- Similarities: Both use a secret key and aim to hide the original message.

Analysis

- Differences and Similarities: Both aim to provide confidentiality using a key, but their core mechanism differs. Substitution changes the identity of the data (A becomes D), while Transposition changes the position of the data (Hello becomes olleH).

- Data Structure & Security Level: Substitution destroys the original data structure but leaves language frequencies (if monoalphabetic). Transposition keeps the original characters (and thus the exact letter frequencies of the plaintext) but destroys the sequence structure.
- Which is more vulnerable to attacks? Substitution (specifically Monoalphabetic) is generally more vulnerable to a ciphertext-only attack because single-letter frequency analysis is incredibly powerful. Transposition ciphers preserve the exact letter frequencies, which immediately tells the attacker it's a transposition cipher, but piecing it back together without the key can be computationally harder if the block size is large.

4. Steganography Concepts



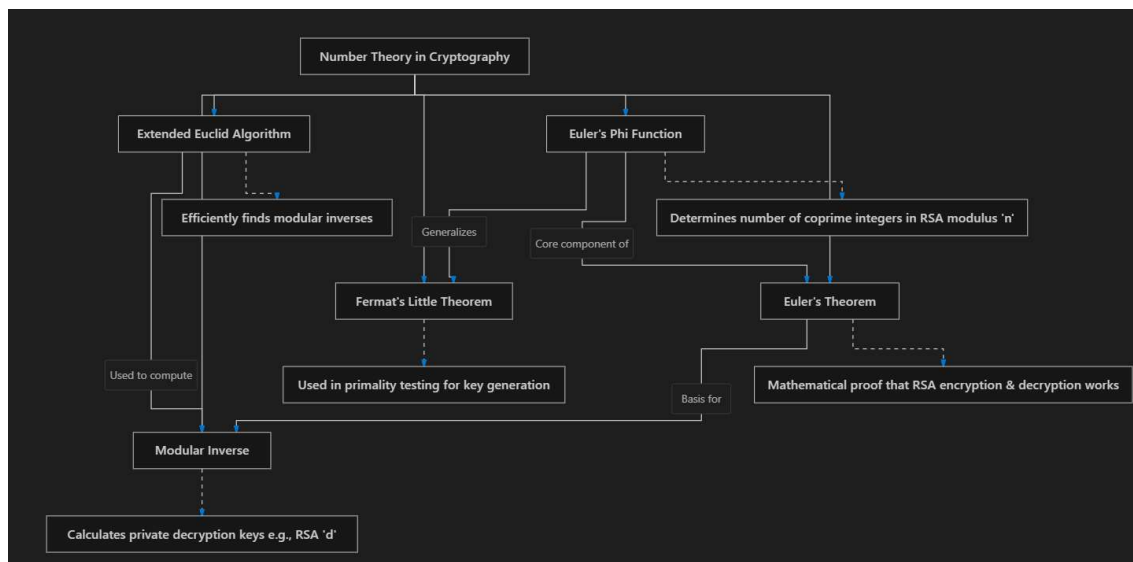
Concept Map

- Data Hiding: Core mechanism. Conceals the very existence of a message.
- Watermarking: Application of Data Hiding. Proves authenticity & copyright protection.
- Survivability: Required for both. Ensures hidden data persists after alterations.

Analysis

- Relationships: Data Hiding is the overarching concept of concealing information. Watermarking is a specific application of data hiding meant to protect intellectual property rather than conceal secret messages. Survivability is a crucial property required for both; the hidden data or watermark must survive benign transformations (like JPEG compression) or malicious attempts to remove it.
- Contribution to Secure Communication: Data Hiding provides security through obscurity. Watermarking ensures integrity and non-repudiation. Survivability ensures the robustness of the secure communication channel against noise and tampering.
- Real-world applications: Data Hiding (Secret comms embedded in an image payload). Watermarking (Invisible digital signature on proprietary photos). Survivability (Hidden metadata surviving image compression/cropping).

5. Number Theory Concepts in Cryptography



Concept Map

- Extended Euclid Algorithm: Used to compute Modular Inverse.
- Euler's Phi Function: Generalizes Fermat's Little Theorem and is the core component of Euler's Theorem.
- Euler's Theorem: Basis for computing Modular Inverse for decryption keys.

Analysis

- Mathematical Connections: The Extended Euclidean Algorithm provides the practical method to find a Modular Inverse. Euler's Phi Function calculates the totient of a number, which is foundational to Euler's Theorem. Euler's Theorem is a generalization of Fermat's Little Theorem.
- Role in Encryption/Decryption: These concepts are the mathematical backbone of RSA public-key encryption. They govern how the public key (e) and private key (d) are generated such that encrypting and then decrypting yields the original message.

Which concept is most critical for public key cryptography?

Euler's Phi Function ($\phi(n)$) & Euler's Theorem are the most critical.

Justification: The entire security of the RSA cryptosystem relies on the fact that computing $\phi(n)$ for a large composite number n (where $n = p \times q$) is computationally infeasible without knowing the prime factors p and q . Euler's Theorem provides the exact mathematical framework that allows the trapdoor function to work.